

Day 2 — Complement to 1000 and Beyond

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute complements to 1000 and 10000 mentally.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: pratiyogi (*modern coinage*); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define complement.** The complement of N to base B is $B - N$. Today we find complements to 1000 mentally.
2. **Worked examples.**
 - $738 \rightarrow 9-7=2, 9-3=6, 10-8=2 \rightarrow \mathbf{262}$. Check: $738 + 262 = 1000$. ✓
 - $405 \rightarrow 9-4=5, 9-0=9, 10-5=5 \rightarrow \mathbf{595}$. Check: $405 + 595 = 1000$. ✓
 - $999 \rightarrow 0, 0, 1 \rightarrow \mathbf{001}$. Edge case: trailing 9s carry differently — “*all from 9 except the LAST non-zero digit, which is from 10.*”
3. **The “trailing zero” caveat.** For 730 (ends in 0): treat the last non-zero digit as the one from 10. $730 \rightarrow 9-7=2, 10-3=7, 0 \rightarrow \mathbf{270}$. Or equivalently: complement 73 to 100 (= 27), append the trailing 0 $\rightarrow 270$.
4. **Pattern algebra.** For $N < 10^k$, the complement is $(10^k - 1) - N + 1 = (10^k - N)$. The “all from 9 last from 10” procedure computes this digit-by-digit.
5. **Practice.** 15-problem mental sprint, complements to 1000 only. 5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Complement to 1000 and Beyond.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).